SPACE SHOOTER GAME PROJECT REPORT

Track Name: Game Development with Python (Pygame)

Setup Instructions:

1. Install Python (version 3.14).
2. Install pygame using: pip install pygame(Community Version)
3. Maintain proper project folder structure with assets and sounds.

Run Instructions:

1. Open terminal in project folder.
2. Run: python main.py
3. Enter player name and click START to begin the game.

Core Gameplay Features:

- Player movement and shooting system
- Enemy spawning and collision detection
- Score system (+10 per enemy)

Boss System:

- Boss appears after certain score
- Multiple boss levels with different images
- Boss health and attack system- Bonus score for defeating boss

Level System:

- Level increases after defeating boss
- Difficulty increases with level
- Level completion shown in Game Over

Player System:

- Lives system (3 hearts)

- Game over when lives reach zero

Power-Up System:

- Random power-up drops

- Health regeneration

UI/UX Features:

- Start screen with background, music, and name input

- HUD showing score, high score, missed enemies

- Game Over screen with stats and buttons

- Game win screen with stas and button.

Sound System:

- Shooting, boss, and death sounds

- Background music

Data Storage:

- High score saved using file/JSON

Visual Effects:

- Background images

- Screen shake effect

- Smooth animations

Advanced Features:

- Multiple boss types

- Restart system

- Modular structure

- OOP implementation

Programming Concepts Used:

- Object-Oriented Programming

- Game loop and event handling

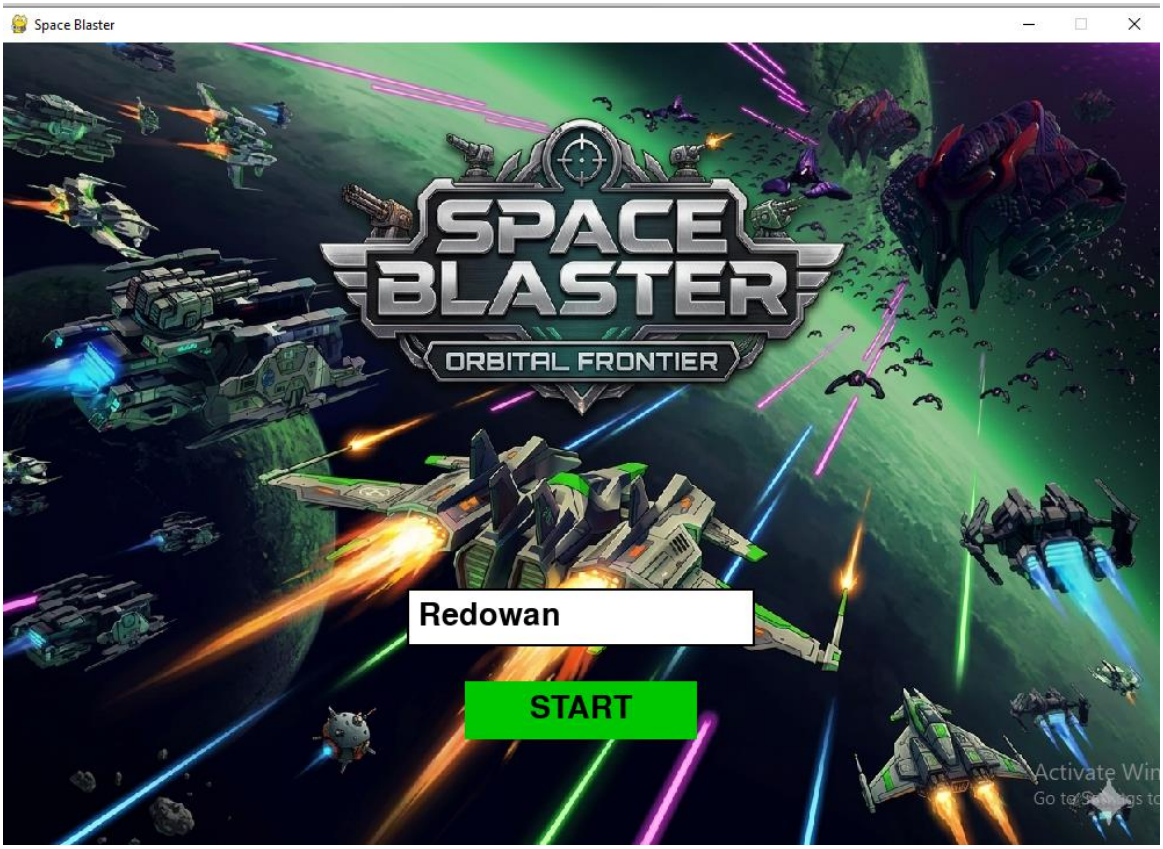
- Collision detection

- File handling

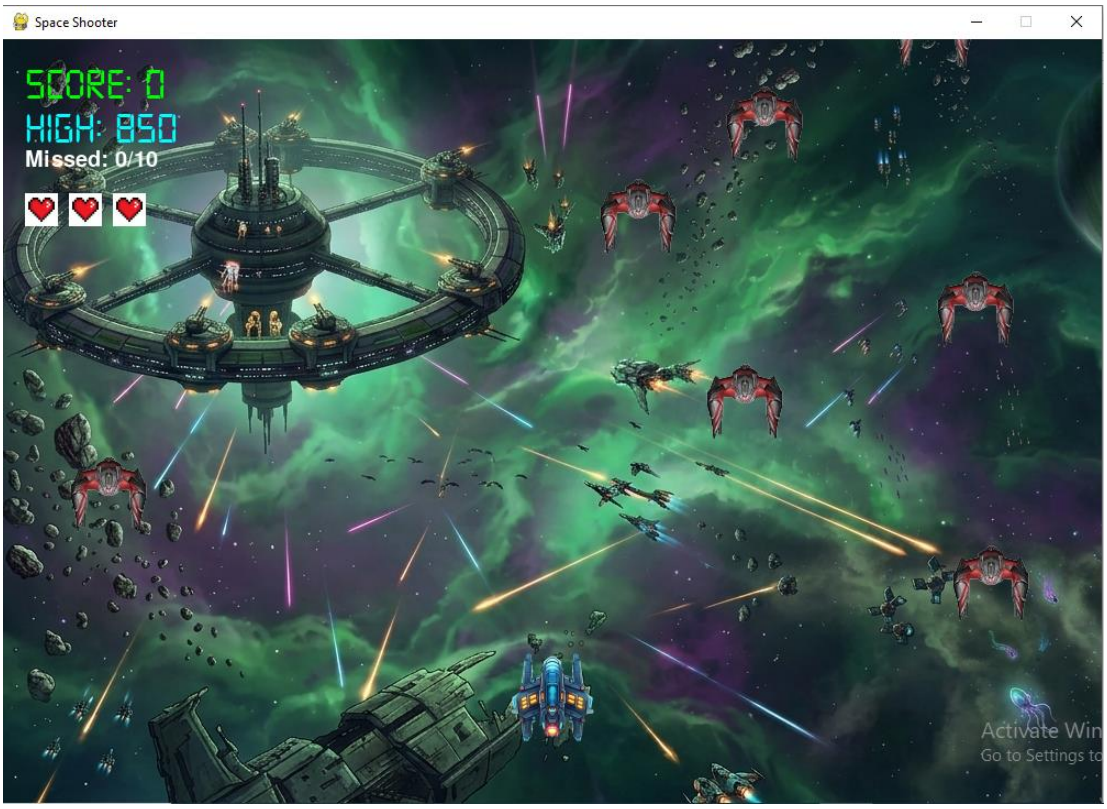
Conclusion:

This project demonstrates a complete 2D game using Pygame, including gameplay mechanics, UI design, sound integration, and data persistence.

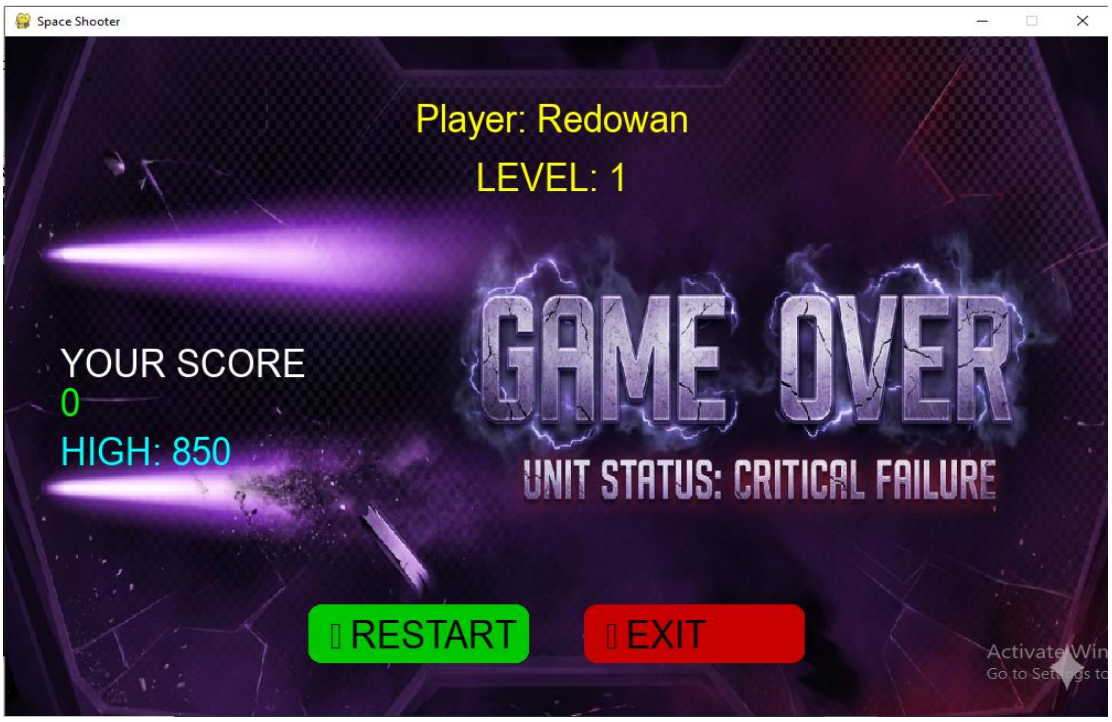
1:



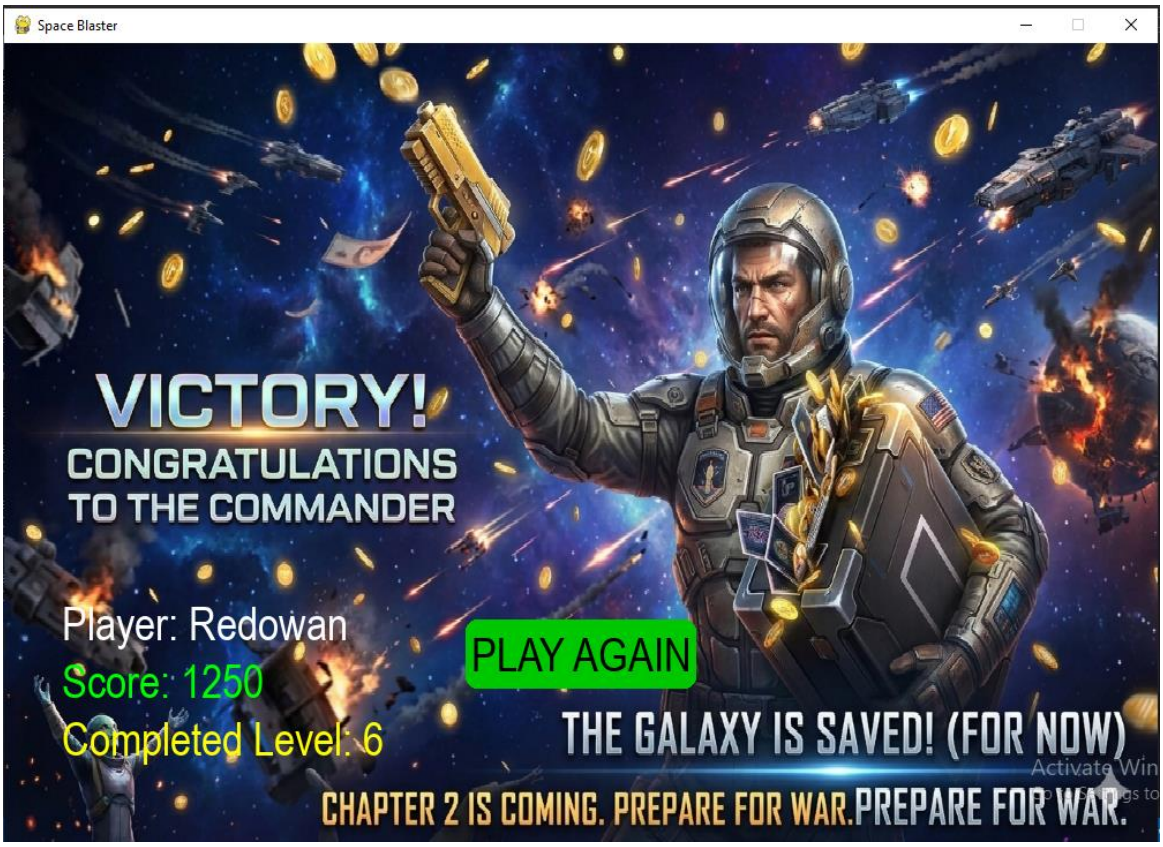
2:


Space Shooter
SCORE: 0
HIGH: 850
Missed: 0/10

3:


Space Shooter
Player: Redowan
LEVEL: 1
YOUR SCORE
0
HIGH: 850
GAME OVER
UNIT STATUS: CRITICAL FAILURE
RESTART
EXIT

4:


Space Blaster
VICTORY!
CONGRATULATIONS
TO THE COMMANDER
Player: Redowan
Score: 1250
Completed Level: 6
PLAY AGAIN
THE GALAXY IS SAVED! (FOR NOW)
CHAPTER 2 IS COMING. PREPARE FOR WAR.PREPARE FOR WAR.